

Phase 5 – Project Development

AI WeatherWise Project

1. Project Setup

- Create the AI Weatherwise project folder and open it in Visual Studio Code.
- Initialize Node.js using `npm init -y`.
- Install Express, Mongoose, bcryptjs, jsonwebtoken, cors, dotenv, multer and @google/genai.
- Install Nodemon as a development dependency.

2. Required Files

- index.js
- .env
- package.json
- .gitignore

3. Environment Configuration

The supplied document specifies PORT, MONGO_URI, JWT_SECRET and GEMINI_API_KEY. Secret values should not be uploaded to a public repository.

4. Backend Development

- Express server initializes middleware and routes.
- Mongoose establishes the MongoDB connection.
- Routing, controllers, middleware, models and utilities are separated.
- JWT authentication and bcrypt password hashing are used.
- Weather data and AI insights are handled through the documented integrations.

5. Development Evidence

- Source code.
- Database/schema files.
- Configuration templates without secret values.
- Working API screenshots.
- README with setup and execution instructions.